

SHARNA HOSSAIN

Software Engineer

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PROFESSIONAL SUMMARY

Software engineer building and delivering high-stakes financial systems end-to-end. Experienced in owning payment and data workflows from complex requirements through production, integrating across systems, and partnering closely with users and stakeholders to deliver practical, reliable solutions. Strong focus on distributed systems, reliability, and shipping quickly in high-impact environments.

WORK EXPERIENCE

Software Engineer II

Morgan Stanley | August 2024 - Present | New York, NY

- Delivered a real-time payments system end-to-end, enabling reliable settlement of high-value transactions and contributing to a broader initiative projected to avoid \$70M in costs
- Designed and implemented distributed system architecture for financial workflows, defining service boundaries and integration patterns to support scalable, maintainable systems
- Partnered with stakeholders across business and engineering to translate complex financial requirements into production-ready systems, aligning on behavior, edge cases, and failure handling
- Built event-driven services enabling reliable communication between internal systems and external partners, improving resilience and reducing operational risk
- Supported rollout of a new real-time trade model, ensuring accurate execution of daily FX transactions at scale
- Modernized a legacy trading platform (GWT → distributed web architecture), decoupling front-end and back-end systems to improve scalability, performance, and long-term maintainability

Pro Bono Developer

Global Poverty Research Lab | March 2026 - Present | Remote

- Enhancing an open-source, Quarto-based data platform by supporting implementation of authentication, role-based access controls, and early interactive analytics within an R/Python stack, improving secure and scalable data access for research and policy users. Part of my role at Morgan Stanley via the Firm's Tech Change Makers program.

Software Engineer I

Morgan Stanley | February 2024 - August 2024 | New York, NY

- Built a microservice supporting real-time settlement workflows, enabling integration between internal systems and external banking partners
- Implemented payment message flows and transaction tracking logic to ensure accuracy and traceability of financial operations
- Produced system design artifacts (ADRs, C4 diagrams) to guide architecture decisions and align cross-functional teams
- Diagnosed and resolved upstream data integrity issues, improving reliability of financial data used across distributed systems

EDUCATION

Coursework in Computer Science | Columbia University | 2022

Associate in Science, Computer Science | Borough of Manhattan Community College | 2021

Coding Bootcamp Certificate | New York Code + Design Academy | 2018

SKILLS & TECHNOLOGIES

Languages: Java, SQL, Python, JavaScript, R

Systems: Distributed Systems, Event-Driven Architecture, Real-Time Payments

Frameworks: Spring Boot, Angular, REST APIs

Tools: Kafka, IBM MQ, Docker, Jenkins, Snowflake, Gradle, Linux, Git